Project 2: Natural Wildfire Ignitions by Ecoregion and Fuel Type

September 28, 2026

0.1 Research Question:

Between 2018 and 2020, does the rate at which lightning strikes convert into wildfire ignitions vary by fuel type across the ecoregions of the contiguous United States, independent of raw lightning exposure?

Revised Research Question: Between 1992 and 2024, how are naturally occuring wildfires distributed across ecoregions and what fuel-types during this time period account for the most natural ignitions?

1 Data:

1. Spatial wildfire occurrence data for the United States, 1992-2020 (7th Edition)(USFS)
   - Federal, State and Local fire data aggreagated
   - Discovery date, final fire size, point location (1 sq mile accuracy), wildfire-cause
   - 2.3 million geo-referenced wildfire records
2. Combined Geostationary Lightning Mapper (GLM) Hourly Thunder Hour Data Product (NASA)
   - 2019 - 2025
   - Two sattelites GOES-18 (west) and GOES-19 (east)
   - Thunder hour: an hour with detected lightning at a location - a frequency-based storm-activity metric, less biased by a sensor's geographic detection-sensitivity variation than raw flash counts
   - Resolution: 5 km
3. LANDFIRE's 40 Scott and Burgan Fire Behavior Fuel Model (FBFM40), CONUS raster (USFS & USGS
   - 40 fuel models classifying surface fuel loading by type (grass, brush, timber, slash), size class, bed depth, and moisture of extinction
   - Fuel assignments aren't static - herbaceous fuels shift between live and dead through the growing season, and burned or otherwise disturbed areas are updated to reflect estimated recovery through 2022
4. Level III Ecoregions of the Conterminous United States (EPA)
   - Ecoregions mark areas of broadly similar ecosystems - a spatial framework for managing and comparing environmental resources across agencies
   - Delineated from patterns across multiple factors (geology, vegetation, climate, soils, land use, wildlife, hydrology), weighted differently region to region
   - Four Levels: Level I (15 regions, North America) → Level II (50) → Level III (85 in the conterminous US) → Level IV (further subdivisions)
5. Backup Wildfire Data: Interagency Fire Occurrence Reporting Modules (InFORM
   - Ongling incorporation of historic data across other agencies is still in progress
   - Data still needs to be backfilled

1.1 Data Processing:

1. CRS
   - Albers Equal Area Conic
     - Preserves area calculation accuracy
     - Minimal distortion for CONUS
2. Zonal Statistics at CONUS Scale for LANDFIRE Raster
   - Windowed (chunked) reads for the raster data set to avoid memory issues in Docker

3. Lightning Data Cropping to CONUS
4. LANDFIRE Attribute Table
    - Translate values to fuel model classes
5. Categorize Fuel Types
    - Single majority class per eco region?
    - Proportions of fuel class per eco region?
    - Lightning strikes per fuel class

1.2 Analysis Methods:

1.2.1 Spatial joins and overlays

- Fire points joined to ecoregion polygons (point-in-polygon)
- Fire points sampled against the fuel raster (tagging each ignition with its fuel class)
- Lightning grid cells overlaid on the fuel raster to assign each cell a majority fuel class
- Statistics of the fuel raster within ecoregion polygons (raster clipped by polygon)

1.2.2 Attribute filtering and aggregation

- Filtering fire records to NWCG_CAUSE == "Natural" and to the date range February 2019–December 2020
- Lightning will use the combined Monthly Thunder Hour product for 2019 and 2020 to summ across CONUS to get a per grid cell count
    - Monthly Thunder Hour: did lightning happen in this cell during this hour, yes or no?
- Aggregating fuel pixel counts within each ecoregion into classes

1.2.3 Choropleth mapping and advanced visualization

- Fuel map colored by their ignition-to-thunder-hour ratio
- Ecoregions colored by their ignition-to-thunder-hour ratio
- Create a map that displays the areas with the most lightning cause wildfire ignitions

[10]:
```
# import libraries

import os
import geopandas as gpd
import pandas as pd
import numpy as np
import rasterio
from IPython.display import display
from rasterio.transform import from_origin
from rasterio.warp import reproject
from rasterio.enums import Resampling
from libpysal import weights
import matplotlib.pyplot as plt


from pathlib import Path
DATA_DIR = Path("data")
```

1.3 Reprojections for Analysis and Geographic Alignment

Projection: EPSG:5070 — NAD83 / Conus Albers Equal Area Conic

- Albers is an equal-area projection that preserves area accuarcy for CONUS

- Minimal distortion for CONUS
- Well-suited for regional comparisons

[2]:
```
# Fire occurrence (7th edition, reprojected to Albers)
# Cache paths so we don't have to recompute everytime we open a the notebook
fire_cache_path = DATA_DIR / "processed/fire_occurrence_7th_albers.gpkg"

if fire_cache_path.exists():
    fire = gpd.read_file(fire_cache_path)
else:
    fire = gpd.read_file(DATA_DIR / "FPA_FOD/FPA_FOD_20260615.gpkg")
    fire = fire.to_crs("EPSG:5070")
    fire.to_file(fire_cache_path, driver="GPKG")

# Fuel type raster (already in EPSG:5070 - no reprojection needed)
fuelClass = rasterio.open(DATA_DIR / "LF2020_FBFM40_CONUS/Tif/LF2020_FBFM40_CONUS.tif")

# Ecoregions (reprojected + dissolved to one polygon per ecoregion)
# Dissolve ecoregions because the downloaded shapefiles split ecoregion polygons at␣
 ↪state-lines
ecoregion_cache_path = DATA_DIR / "processed/ecoregions_l3_albers_nb.gpkg"

if ecoregion_cache_path.exists():
    ecoregion = gpd.read_file(ecoregion_cache_path)
else:
    ecoregion = gpd.read_file(DATA_DIR / "ecoregions/us_eco_l3_state_boundaries.shp")
    ecoregion = ecoregion.to_crs("EPSG:5070")
    # Dissolve on the ecoregion identifier
    ecoregion = ecoregion.dissolve(by="US_L3CODE", as_index=False)
    ecoregion.to_file(ecoregion_cache_path, driver="GPKG")
```

[3]:
```
# Look at the datasets to identify what needs to be cleaned
print("Fire Occurrence")
pd.set_option('display.max_columns', None)
display(fire.head(10))

print("Ecoregions")
display(ecoregion.head(10))

print("Fuel Model Classes (LANDFIRE attribute table)")
fuel_lookup = pd.read_csv(DATA_DIR / "LF2020_FBFM40_CONUS/CSV_Data/LF2020_FBFM40.csv")
display(fuel_lookup.head(10))
```

```
Fire Occurrence
```

	FOD_ID	FPA_ID	SOURCE_SYSTEM_TYPE	SOURCE_SYSTEM	NWCG_REPORTING_AGENCY	\
0	1	FS-1418826	FED	FS-FIRESTAT	FS	
1	2	FS-1418827	FED	FS-FIRESTAT	FS	
2	3	FS-1418835	FED	FS-FIRESTAT	FS	
3	4	FS-1418845	FED	FS-FIRESTAT	FS	
4	5	FS-1418847	FED	FS-FIRESTAT	FS	
5	6	FS-1418849	FED	FS-FIRESTAT	FS	
6	7	FS-1418851	FED	FS-FIRESTAT	FS	
7	8	FS-1418854	FED	FS-FIRESTAT	FS	

```
8       9  FS-1418856               FED   FS-FIRESTAT                    FS
9      10  FS-1418859               FED   FS-FIRESTAT                    FS

  NWCG_REPORTING_UNIT_ID        NWCG_REPORTING_UNIT_NAME  \
0                 USCAPNF          Plumas National Forest
1                 USCAENF        Eldorado National Forest
2                 USCAENF        Eldorado National Forest
3                 USCAENF        Eldorado National Forest
4                 USCAENF        Eldorado National Forest
5                 USCAENF        Eldorado National Forest
6                 USCAENF        Eldorado National Forest
7                 USCASHF  Shasta-Trinity National Forest
8                 USCASHF  Shasta-Trinity National Forest
9                 USCAENF        Eldorado National Forest

  SOURCE_REPORTING_UNIT      SOURCE_REPORTING_UNIT_NAME LOCAL_FIRE_REPORT_ID  \
0                  0511          Plumas National Forest                    1
1                  0503        Eldorado National Forest                   13
2                  0503        Eldorado National Forest                   27
3                  0503        Eldorado National Forest                   43
4                  0503        Eldorado National Forest                   44
5                  0503        Eldorado National Forest                   54
6                  0503        Eldorado National Forest                   58
7                  0514  Shasta-Trinity National Forest                    3
8                  0514  Shasta-Trinity National Forest                    5
9                  0503        Eldorado National Forest                   61

  LOCAL_INCIDENT_ID FIRE_CODE   FIRE_NAME ICS_209_PLUS_INCIDENT_JOIN_ID  \
0            PNF-47      BJ8K    FOUNTAIN                         None
1                13      AAC0      PIGEON                         None
2               021      A32W       SLACK                         None
3                 6      None        DEER                         None
4                 7      None    STEVENOT                         None
5                 8      None      HIDDEN                         None
6                 9      None        FORK                         None
7                02      BK5X       SLATE                         None
8                03      BLPQ      SHASTA                         None
9                10      None  TANGLEFOOT                         None

  ICS_209_PLUS_COMPLEX_JOIN_ID MTBS_ID MTBS_FIRE_NAME COMPLEX_NAME  FIRE_YEAR  \
0                         None    None           None         None       2005
1                         None    None           None         None       2004
2                         None    None           None         None       2004
3                         None    None           None         None       2004
4                         None    None           None         None       2004
5                         None    None           None         None       2004
6                         None    None           None         None       2004
7                         None    None           None         None       2005
8                         None    None           None         None       2005
9                         None    None           None         None       2004

            DISCOVERY_DATE  DISCOVERY_DOY DISCOVERY_TIME  \
0 2005-02-02 00:00:00+00:00             33           1300
```

```
1 2004-05-12 00:00:00+00:00            133           0845
2 2004-05-31 00:00:00+00:00            152           1921
3 2004-06-28 00:00:00+00:00            180           1600
4 2004-06-28 00:00:00+00:00            180           1600
5 2004-06-30 00:00:00+00:00            182           1800
6 2004-07-01 00:00:00+00:00            183           1800
7 2005-03-08 00:00:00+00:00             67           1300
8 2005-03-15 00:00:00+00:00             74           1200
9 2004-07-01 00:00:00+00:00            183           1800

  NWCG_CAUSE_CLASSIFICATION                         NWCG_GENERAL_CAUSE  \
0                     Human  Power generation/transmission/distribution
1                   Natural                                     Natural
2                     Human                   Debris and open burning
3                   Natural                                     Natural
4                   Natural                                     Natural
5                   Natural                                     Natural
6                   Natural                                     Natural
7                     Human                   Debris and open burning
8                     Human                   Debris and open burning
9                   Natural                                     Natural

  NWCG_CAUSE_AGE_CATEGORY                CONT_DATE  CONT_DOY CONT_TIME  \
0                    None 2005-02-02 00:00:00+00:00      33.0      1730
1                    None 2004-05-12 00:00:00+00:00     133.0      1530
2                    None 2004-05-31 00:00:00+00:00     152.0      2024
3                    None 2004-07-03 00:00:00+00:00     185.0      1400
4                    None 2004-07-03 00:00:00+00:00     185.0      1200
5                    None 2004-07-01 00:00:00+00:00     183.0      1600
6                    None 2004-07-02 00:00:00+00:00     184.0      1400
7                    None 2005-03-08 00:00:00+00:00      67.0      1600
8                    None 2005-03-15 00:00:00+00:00      74.0      1700
9                    None 2004-07-02 00:00:00+00:00     184.0      1800

   FIRE_SIZE FIRE_SIZE_CLASS   LATITUDE   LONGITUDE       OWNER_DESCR STATE  \
0       0.10               A  40.036944 -121.005833              USFS    CA
1       0.25               A  38.933056 -120.404444              USFS    CA
2       0.10               A  38.984167 -120.735556  STATE OR PRIVATE    CA
3       0.10               A  38.559167 -119.913333              USFS    CA
4       0.10               A  38.559167 -119.933056              USFS    CA
5       0.10               A  38.635278 -120.103611              USFS    CA
6       0.10               A  38.688333 -120.153333              USFS    CA
7       0.80               B  40.968056 -122.433889  STATE OR PRIVATE    CA
8       1.00               B  41.233611 -122.283333  STATE OR PRIVATE    CA
9       0.10               A  38.548333 -120.149167              USFS    CA

  COUNTY FIPS_CODE         FIPS_NAME                          geometry
0     63     06063     Plumas County  POINT (-2090649.206 2167768.542)
1     61     06061     Placer County  POINT (-2072914.128 2035040.471)
2     17     06017  El Dorado County  POINT (-2098930.807 2047846.468)
3      3     06003     Alpine County  POINT (-2042526.985 1983845.809)
4      3     06003     Alpine County  POINT (-2044175.784 1984269.866)
5      5     06005     Amador County  POINT (-2056290.102 1996209.432)
```

```
6     17     06017  El Dorado County  POINT (-2058945.094 2003042.150)
7   None      None              None  POINT (-2178397.193 2300347.756)
8   None      None              None  POINT (-2158190.184 2325509.048)
9      5     06005     Amador County  POINT (-2062542.378 1987763.701)
```

Ecoregions

```
  US_L3CODE               US_L3NAME NA_L3CODE  \
0         1             Coast Range     7.1.8
1        10         Columbia Plateau    10.1.2
2        11           Blue Mountains     6.2.9
3        12        Snake River Plain    10.1.8
4        13  Central Basin and Range    10.1.5
5        14   Mojave Basin and Range    10.2.1
6        15         Northern Rockies     6.2.3
7        16          Idaho Batholith    6.2.15
8        17           Middle Rockies    6.2.10
9        18            Wyoming Basin    10.1.4

                           NA_L3NAME NA_L2CODE                 NA_L2NAME  \
0                        Coast Range       7.1  MARINE WEST COAST FOREST
1                      Columbia Plateau      10.1              COLD DESERTS
2                        Blue Mountains       6.2        WESTERN CORDILLERA
3                     Snake River Plain      10.1              COLD DESERTS
4               Central Basin and Range      10.1              COLD DESERTS
5                Mojave Basin and Range      10.2              WARM DESERTS
6  Columbia Mountains/Northern Rockies       6.2        WESTERN CORDILLERA
7                       Idaho Batholith       6.2        WESTERN CORDILLERA
8                        Middle Rockies       6.2        WESTERN CORDILLERA
9                         Wyoming Basin      10.1              COLD DESERTS

  NA_L1CODE                        NA_L1NAME  STATE_NAME  EPA_REGION  \
0         7         MARINE WEST COAST FOREST  California           9
1        10           NORTH AMERICAN DESERTS       Idaho          10
2         6  NORTHWESTERN FORESTED MOUNTAINS       Idaho          10
3        10           NORTH AMERICAN DESERTS       Idaho          10
4        10           NORTH AMERICAN DESERTS  California           9
5        10           NORTH AMERICAN DESERTS     Arizona           9
6         6  NORTHWESTERN FORESTED MOUNTAINS       Idaho          10
7         6  NORTHWESTERN FORESTED MOUNTAINS       Idaho          10
8         6  NORTHWESTERN FORESTED MOUNTAINS       Idaho          10
9        10           NORTH AMERICAN DESERTS    Colorado           8

                      L3_KEY                         L2_KEY  \
0               1  Coast Range  7.1  MARINE WEST COAST FOREST
1         10  Columbia Plateau            10.1  COLD DESERTS
2           11  Blue Mountains        6.2  WESTERN CORDILLERA
3        12  Snake River Plain            10.1  COLD DESERTS
4  13  Central Basin and Range            10.1  COLD DESERTS
5   14  Mojave Basin and Range            10.2  WARM DESERTS
6         15  Northern Rockies        6.2  WESTERN CORDILLERA
7          16  Idaho Batholith        6.2  WESTERN CORDILLERA
8           17  Middle Rockies        6.2  WESTERN CORDILLERA
9            18  Wyoming Basin            10.1  COLD DESERTS
```

```
                              L1_KEY  \
0         7  MARINE WEST COAST FOREST
1          10  NORTH AMERICAN DESERTS
2  6  NORTHWESTERN FORESTED MOUNTAINS
3          10  NORTH AMERICAN DESERTS
4          10  NORTH AMERICAN DESERTS
5          10  NORTH AMERICAN DESERTS
6  6  NORTHWESTERN FORESTED MOUNTAINS
7  6  NORTHWESTERN FORESTED MOUNTAINS
8  6  NORTHWESTERN FORESTED MOUNTAINS
9          10  NORTH AMERICAN DESERTS

                                             geometry
0  MULTIPOLYGON (((-2326956.693 1961322.753, -232…
1  MULTIPOLYGON (((-1528416.035 2731696.232, -152…
2  POLYGON ((-1672022.952 2558457.555, -1672093.0…
3  POLYGON ((-1202707.144 2395971.238, -1202710.6…
4  POLYGON ((-1858634.567 1762679.205, -1859348.8…
5  POLYGON ((-1579497.947 1659182.070, -1579385.6…
6  POLYGON ((-1423929.253 2779186.628, -1424035.9…
7  POLYGON ((-1423805.863 2779074.161, -1423720.2…
8  MULTIPOLYGON (((-1215843.351 2390715.034, -121…
9  POLYGON ((-948977.546 2056277.979, -948857.610…
```

Fuel Model Classes (LANDFIRE attribute table)

	VALUE	FBFM40	R	G	B	RED	GREEN	BLUE
0	-9999	Fill-NoData	255	255	255	1.000000	1.000000	1.000000
1	91	NB1	104	104	104	0.407843	0.407843	0.407843
2	92	NB2	225	225	225	0.882353	0.882353	0.882353
3	93	NB3	255	237	237	1.000000	0.929412	0.929412
4	98	NB8	0	14	214	0.000000	0.054902	0.839216
5	99	NB9	77	110	112	0.301961	0.431373	0.439216
6	101	GR1	255	235	190	1.000000	0.921569	0.745098
7	102	GR2	255	211	115	1.000000	0.827451	0.450980
8	103	GR3	255	236	139	1.000000	0.925490	0.545098
9	104	GR4	255	255	115	1.000000	1.000000	0.450980

[4]:
```
# Check each dataset for empty/NaN values
print(fire.isnull().sum())
print(ecoregion.isnull().sum())
print(fuel_lookup.isnull().sum())
```

```
FOD_ID                          0
FPA_ID                          0
SOURCE_SYSTEM_TYPE              0
SOURCE_SYSTEM                   0
NWCG_REPORTING_AGENCY           0
NWCG_REPORTING_UNIT_ID          0
NWCG_REPORTING_UNIT_NAME        0
SOURCE_REPORTING_UNIT           0
SOURCE_REPORTING_UNIT_NAME      0
LOCAL_FIRE_REPORT_ID      2183789
```

```
LOCAL_INCIDENT_ID                749700
FIRE_CODE                       2201851
FIRE_NAME                        974876
ICS_209_PLUS_INCIDENT_JOIN_ID   2617279
ICS_209_PLUS_COMPLEX_JOIN_ID    2655444
MTBS_ID                         2645277
MTBS_FIRE_NAME                  2645276
COMPLEX_NAME                    2654373
FIRE_YEAR                             0
DISCOVERY_DATE                        0
DISCOVERY_DOY                         0
DISCOVERY_TIME                   829725
NWCG_CAUSE_CLASSIFICATION             0
NWCG_GENERAL_CAUSE                    0
NWCG_CAUSE_AGE_CATEGORY         2582371
CONT_DATE                        997216
CONT_DOY                         997216
CONT_TIME                       1122332
FIRE_SIZE                             0
FIRE_SIZE_CLASS                       0
LATITUDE                              0
LONGITUDE                             0
OWNER_DESCR                           0
STATE                                 0
COUNTY                           667699
FIPS_CODE                        667699
FIPS_NAME                        667699
geometry                              0
dtype: int64
US_L3CODE     0
US_L3NAME     0
NA_L3CODE     0
NA_L3NAME     0
NA_L2CODE     0
NA_L2NAME     0
NA_L1CODE     0
NA_L1NAME     0
STATE_NAME    0
EPA_REGION    0
L3_KEY        0
L2_KEY        0
L1_KEY        0
geometry      0
dtype: int64
VALUE     0
FBFM40    0
R         0
G         0
B         0
RED       0
GREEN     0
BLUE      0
dtype: int64
```

[5]:
```
# Simplify the Fire Occurance Dataset to suit the needs of the analysis
# Cleaned data is free of empty/NaN values
fire_clean = fire[[
    "FIRE_YEAR", "DISCOVERY_DATE", "NWCG_CAUSE_CLASSIFICATION",
    "NWCG_GENERAL_CAUSE", "FIRE_SIZE", "STATE", "geometry"
]].copy()

# Look at new dataframe
display(fire_clean.head(10))
```

```
   FIRE_YEAR            DISCOVERY_DATE NWCG_CAUSE_CLASSIFICATION  \
0       2005 2005-02-02 00:00:00+00:00                      Human
1       2004 2004-05-12 00:00:00+00:00                    Natural
2       2004 2004-05-31 00:00:00+00:00                      Human
3       2004 2004-06-28 00:00:00+00:00                    Natural
4       2004 2004-06-28 00:00:00+00:00                    Natural
5       2004 2004-06-30 00:00:00+00:00                    Natural
6       2004 2004-07-01 00:00:00+00:00                    Natural
7       2005 2005-03-08 00:00:00+00:00                      Human
8       2005 2005-03-15 00:00:00+00:00                      Human
9       2004 2004-07-01 00:00:00+00:00                    Natural

                          NWCG_GENERAL_CAUSE  FIRE_SIZE STATE  \
0  Power generation/transmission/distribution       0.10    CA
1                                    Natural       0.25    CA
2                     Debris and open burning       0.10    CA
3                                    Natural       0.10    CA
4                                    Natural       0.10    CA
5                                    Natural       0.10    CA
6                                    Natural       0.10    CA
7                     Debris and open burning       0.80    CA
8                     Debris and open burning       1.00    CA
9                                    Natural       0.10    CA

                          geometry
0  POINT (-2090649.206 2167768.542)
1  POINT (-2072914.128 2035040.471)
2  POINT (-2098930.807 2047846.468)
3  POINT (-2042526.985 1983845.809)
4  POINT (-2044175.784 1984269.866)
5  POINT (-2056290.102 1996209.432)
6  POINT (-2058945.094 2003042.150)
7  POINT (-2178397.193 2300347.756)
8  POINT (-2158190.184 2325509.048)
9  POINT (-2062542.378 1987763.701)
```

[6]:
```
fire_fuel_cache_path = DATA_DIR / "processed/natural_fire_fuel_ecoregion.gpkg"

if fire_fuel_cache_path.exists():
    fire_fuel = gpd.read_file(fire_fuel_cache_path)
# Only look at naturally caused fires
else:
```

```
    natural_fire = fire[fire["NWCG_CAUSE_CLASSIFICATION"] == "Natural"].copy()

    # Build a list of x,y pairs of point geom for natural fires
    coords = [(geom.x, geom.y) for geom in natural_fire.geometry]
    # Look up the rasters pixel by coords (val[0] removes the raterio wrapper and returns␣
 ↪an int to the new col FUEL_CLASS to natural_fire
    natural_fire["FUEL_CLASS"] = [val[0] for val in fuelClass.sample(coords)]

    # Spatial join tags each fire WITHIN its containing ecoregion
    # geopandas spatial join matches rows based on geom
    fire_fuel = gpd.sjoin(natural_fire, ecoregion, how="left", predicate="within")

    fire_fuel.to_file(fire_fuel_cache_path, driver="GPKG")

# Clip fire points to the lower 48
# The fire data covers all of the US, but the ecoregions only cover the lower 48.
# The left join in the fire_fuel cell kept every fire, so fires outside the
# ecoregions have NaN in the ecoregion columns. Keeping only rows with an ecoregion name.
fire_fuel = fire_fuel[fire_fuel["US_L3NAME"].notna()].copy()

# Raw ignition counts per ecoregion / fuel type combination
counts = fire_fuel.groupby(["US_L3NAME", "FUEL_CLASS"]).size().
 ↪reset_index(name="ignition_count")
print(counts)
```

```
                    US_L3NAME  FUEL_CLASS  ignition_count
0     Acadian Plains and Hills          91              37
1     Acadian Plains and Hills          98              12
2     Acadian Plains and Hills         101               4
3     Acadian Plains and Hills         102              19
4     Acadian Plains and Hills         103               8
…                          …         …              …
1730             Wyoming Basin         183             142
1731             Wyoming Basin         184               5
1732             Wyoming Basin         185               2
1733             Wyoming Basin         186              42
1734             Wyoming Basin         188              39

[1735 rows x 3 columns]
```

[7]:
```
# ---- Majority fuel class per grid cell ----
# Builds a coarse grid over CONUS where each cell stores the most common fuel class
# found inside it, so density values can later be grouped by fuel class.

cell_size = 10_000  # grid cell width/height in meters (EPSG:5070 units) - I don't know if␣
 ↪this is the right size...we could definitely adjust this

# Grid extent = bounding box of the ecoregions (i.e., CONUS)
xmin, ymin, xmax, ymax = ecoregion.total_bounds

# Columns/rows needed to cover that box; ceil so the far edges aren't cut off
ncols = int(np.ceil((xmax - xmin) / cell_size))
```

```
nrows = int(np.ceil((ymax - ymin) / cell_size))

# Affine transform: ties grid row/column positions to real-world coordinates.
grid_transform = from_origin(xmin, ymax, cell_size, cell_size)

# Cell size is in the filename, so changing it never reuses a stale cached array
fuel_mode_cache_path = DATA_DIR / f"processed/fuel_mode_{cell_size // 1000}km.npy"

if fuel_mode_cache_path.exists():
    fuel_mode = np.load(fuel_mode_cache_path)  # reuse the saved array
else:
    # Empty grid to fill, pre-set to the nodata value
    fuel_mode = np.full((nrows, ncols), -9999, dtype="int16")

    # Resample the 30 m fuel raster onto the coarse grid.
    # Resampling.mode: each cell gets the most common fuel code among the pixels inside it
    reproject(
        source=rasterio.band(fuelClass, 1),  # band 1 = fuel model codes
        destination=fuel_mode,               # array the results are written into
        dst_transform=grid_transform,        # where the grid sits in the world
        dst_crs="EPSG:5070",                 # same CRS as the source: regridding only, no␣
 ↪reprojection
        src_nodata=-9999,                    # ignore nodata pixels in the source
        dst_nodata=-9999,                    # cells with no valid pixels stay -9999
        resampling=Resampling.mode,
    )
    np.save(fuel_mode_cache_path, fuel_mode)  # slow step runs once, then it's cached
```

1.4 Where we are at:

Spatial Joins and Overlays

- fire points joined to ecoregion polygons, fuel raster sampled at each fire, fuel raster resampled onto the 10 km grid
- To Do: overlay the density grid on the fuel grid, then draw ecoregion boundaries over the surface

Attribute filtering and aggregation

- filtered to natural fires, dissolved ecoregions by code, ignition counts per ecoregion and fuel class, majority fuel class per grid cell
- To Do: mean density per fuel class. Normalizing counts by how much of each fuel exists?

Choropleth and advanced visualization

- To Do: choropleth of ignition counts (or the normalized rate) per ecoregion
- To Do: density surface with ecoregion outlines on top
- To Do (optional): one density map per fuel class

Additionally: How do we want to use patial weights (queen contiguity weights)? I was thinking we could compare compare fire occurrence across ecoregions, then use spatial weights (and maybe Moran's I) to find clusters of neighboring ecoregions with similarly high or low ignition rates. Thoughts? (see below).

[8]:
```
# Neighbors = ecoregions
w = weights.Queen.from_dataframe(ecoregion)
```

```
w.transform = "R"   # Row-standardizing divides each weight by that row's total, so every␣
 ↪row sums to 1

print("Ecoregions:", w.n)
print("Neighbors per ecoregion (min / mean / max):",
      w.min_neighbors, round(w.mean_neighbors, 1), w.max_neighbors)
print("Islands (no neighbors):", w.islands)
```

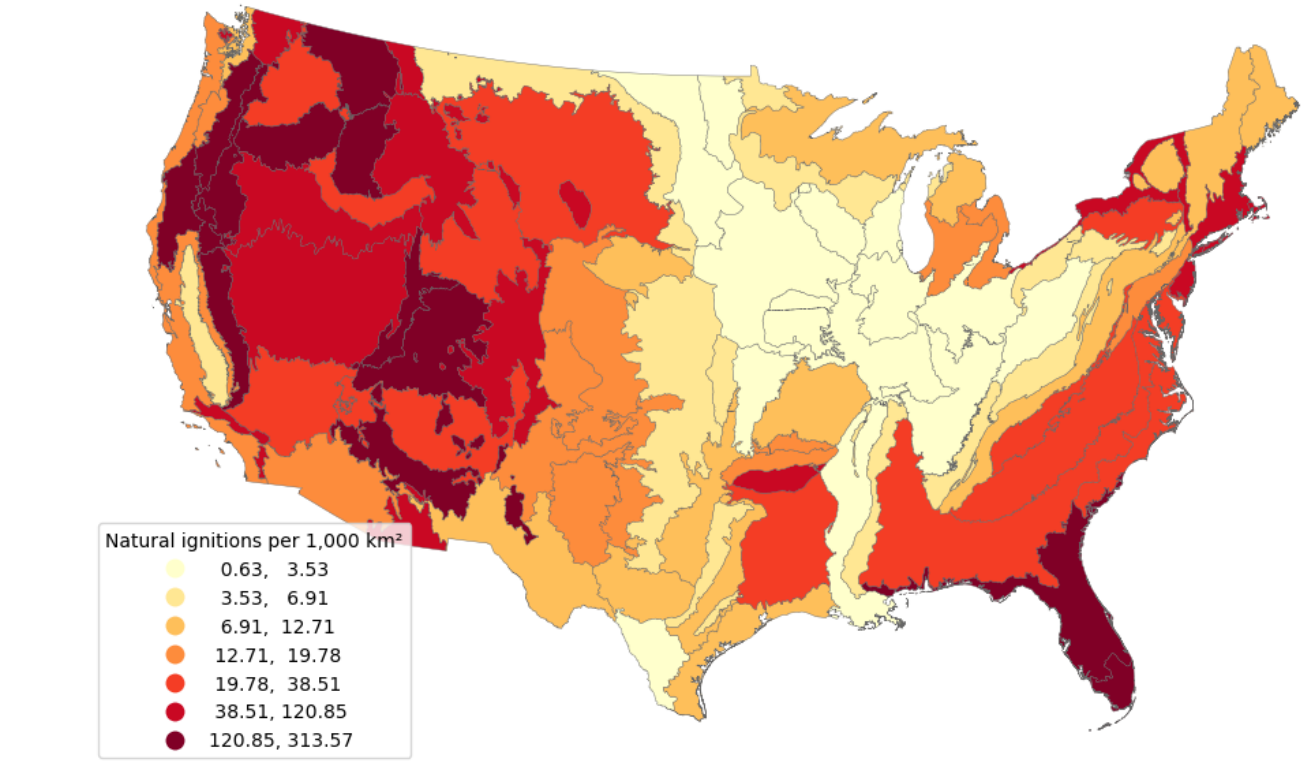
```
/tmp/ipykernel_4516/2281662194.py:2: FutureWarning: `use_index` defaults to
False but will default to True in future. Set True/False directly to control
this behavior and silence this warning
  w = weights.Queen.from_dataframe(ecoregion)
```

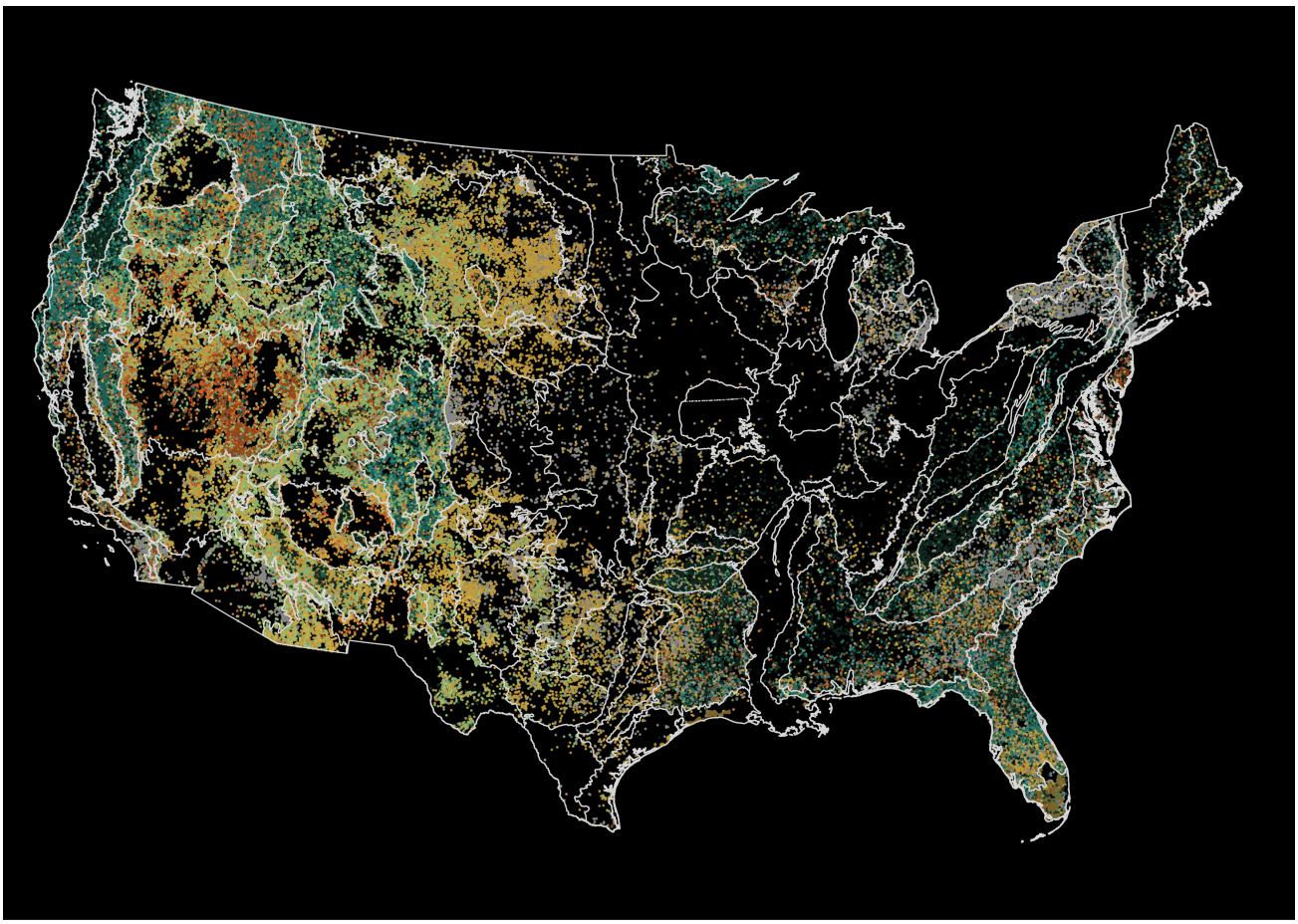
```
Ecoregions: 85
Neighbors per ecoregion (min / mean / max): 1 5.4 10
Islands (no neighbors): []
```

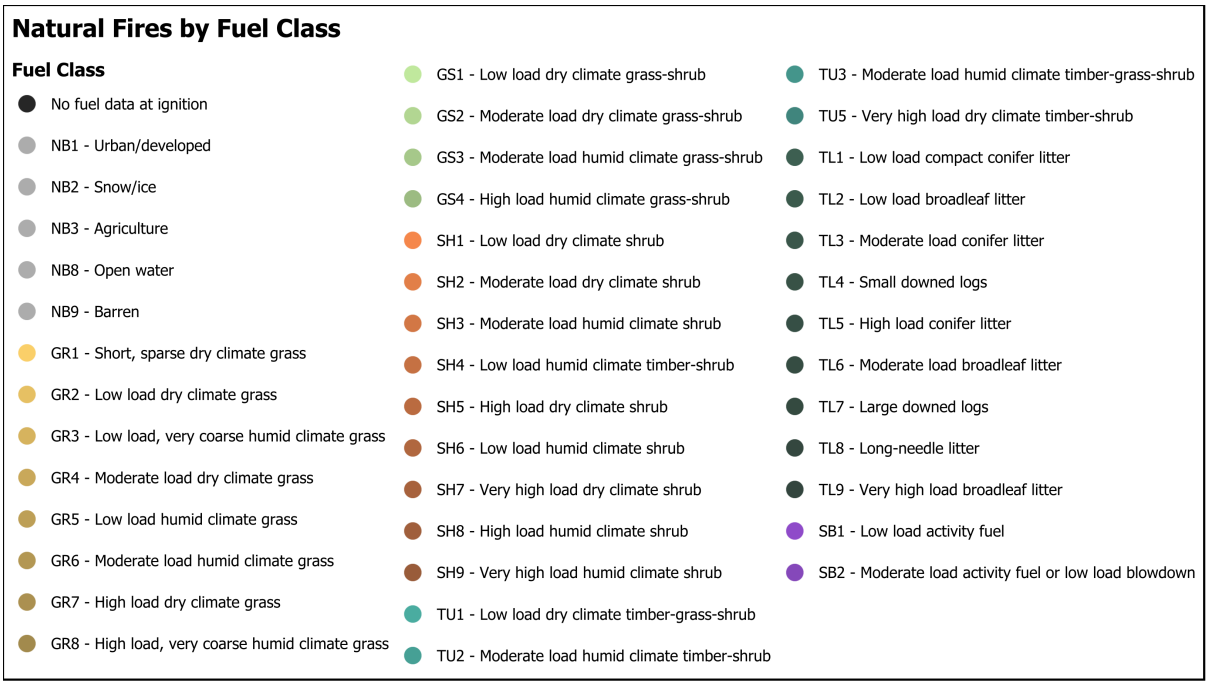
[11]:
```
# Natural ignitions per ecoregion
ignitions = fire_fuel.groupby("US_L3CODE").size()
# Create a new col of ignitions per ecoregion
ecoregion["ignitions"] = ecoregion["US_L3CODE"].map(ignitions).fillna(0)

# .area is in meters squared - divide by 1e9 to get per 1,000 km squared
# Create a new col that contains the normalized ignition data
ecoregion["ign_per_1000km2"] = ecoregion["ignitions"] / (ecoregion.area / 1e9)

# Plot the data
fig, ax = plt.subplots(figsize=(12, 8))
ecoregion.plot(column="ign_per_1000km2", cmap="YlOrRd", scheme="quantiles", k=7,
               edgecolor="0.4", linewidth=0.3, legend=True,
               legend_kwds={"title": "Natural ignitions per 1,000 km²", "loc": "lower␣
 ↪left"},
               ax=ax)
ax.set_title("Natural wildfire ignitions per 1,000 km² by ecoregion, 1992-2024")
ax.set_axis_off()
plt.show()
```


Natural wildfire ignitions per 1,000 km² by ecoregion, 1992–2024
Natural ignitions per 1,000 km²
0.63, 3.53
3.53, 6.91
6.91, 12.71
12.71, 19.78
19.78, 38.51
38.51, 120.85
120.85, 313.57





[9]:
```
# ============================================================
# LIGHTNING DATA - on hold and ready to uncomment when revisited
# ============================================================

# --- Acquisition (already run once; files already exist locally) ---
# %pip install earthaccess
# import earthaccess

# earthaccess.login(strategy="interactive", persist=True)

# results = earthaccess.search_data(
#     short_name="glmmth",
#     temporal=("2019-01-01", "2024-12-31"),
# )
# files = earthaccess.download(results, "work/Project2/data")

# --- Processing: crop each file to CONUS, sum 12 months, sum both years ---
# import glob
# import xarray as xr

# files = sorted(glob.glob(str(DATA_DIR / "glmmth/GLMc_th_*.nc")))

# def load_and_crop(path):
#     ds = xr.open_dataset(path)
#     ds = ds.assign_coords(lat=("nlat", ds.lat.values), lon=("nlon", ds.lon.values))
#     ds = ds.swap_dims({"nlat": "lat", "nlon": "lon"})
```

```
#     lat_vals = ds.lat.values
#     lat_slice = slice(24, 50) if lat_vals[0] < lat_vals[-1] else slice(50, 24)
#     return ds.sel(lat=lat_slice, lon=slice(-125, -66))

# datasets = [load_and_crop(f) for f in files]
# lightning = sum(ds.thunder_hours.sum(dim="nmon") for ds in datasets)
```